Spring 1988

Problem 1 (Sp88). Suppose that $f(x)$, $-\infty < x < \infty$, is a continuous real valued function, that $f'(x)$ exists for $x \neq 0$, and that $\lim_{x\to 0} f'(x)$ exists. Prove that $f'(0)$ exists.

Problem 2 (Sp88, Sp96, Sp10). Determine the rightmost decimal digit of $A = 17^{17^{17}}$.

Problem 3 (Sp88, Sp96). If a finite homogeneous system of linear equations with rational coefficients has a nontrivial complex solution, need it have a nontrivial rational solution? Give a proof or a counterexample.

Problem 4 (Sp88). True or false: A function $f(z)$ analytic on $|z-a| < r$ and continuous on $|z-a| \leqslant r$ extends, for some $\delta > 0$, to a function analytic on $|z-a| < r+\delta$? Give a proof or a counterexample.

Problem 5 (Sp88). Let D be a group of order $2n$, where n is odd, with a subgroup H of order n satisfying $xhx^{-1} = h^{-1}$ for all h in H and all x in $D \setminus H$. Prove that H is commutative and that every element of $D \setminus H$ is of order 2.

Problem 6 (Sp88). Prove or disprove: There is a real $n \times n$ matrix A such that $A^2 + 2A + 5I = 0$ if and only if n is even.

Problem 7 (Sp88). Let R be a commutative ring with identity and $a \in R$. Let n and m be positive integers, and write $d = \gcd\{n, m\}$. Prove that the ideal of R generated by $a^n - 1$ and $a^m - 1$ is the same as the ideal generated by $a^d - 1$.

Problem 8 (Sp88, Fa13). For $a > 1$ and $n = 0, 1, 2, \ldots$, evaluate the integrals

$$C_n(a) = \int_{-\pi}^{\pi} \frac{\cos n\theta}{a - \cos\theta}\,\mathrm{d}\theta \qquad S_n(a) = \int_{-\pi}^{\pi} \frac{\sin n\theta}{a - \cos\theta}\,\mathrm{d}\theta$$

Problem 9 (Sp88). Prove that the integrals

$$\int_0^\infty \cos x^2\,\mathrm{d}x \qquad \text{and} \qquad \int_0^\infty \sin x^2\,\mathrm{d}x$$

converge.

Problem 10 (Sp88). 1. Let U be an open connected subset of the complex plane, f an analytic function in U, not identically 0, and n a positive integer. Assume that f has an analytic n-th root in U; that is, there is an analytic function g in U such that $g^n = f$. Prove that f has exactly n analytic n-th roots in U.

2. Give an example of a continuous real valued function on [0, 1] that has more than two continuous square roots on [0, 1].

Problem 11 (Sp88, Sp01). Let $1 \in S_9$ be the identity permutation. Determine the minimum of all positive integers m such that every $\sigma \in S_9$ satisfies $\sigma^m = 1$. Determine also the minimum of all positive integers m such that every $\sigma \in A_9$ satisfies $\sigma^m = 1$.

Problem 12 (Sp88). For each real value of the parameter t, determine the number of real zeros, counting multiplicities, of the cubic polynomial $p_t(x) = \left(1+t^2\right)x^3 - 3t^3x + t^4$.

Problem 13 (Fa79, Sp88, Fa91). Prove that every finite group of order > 2 has a nontrivial automorphism.

Problem 14 (Sp88). Let the function f be analytic in the open unit disc of the complex plane and real valued on the radii $[0, 1)$ and $[0, e^{i\pi\sqrt{2}})$. Prove that f is constant.

Problem 15 (Sp88, Sp14). Compute A^{10} for the matrix

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ -1 & -1 & 1 \end{pmatrix}$$

Problem 16 (Sp88, Fa19). Let X be a set and V a real vector space of real valued functions on X of dimension n, $0 < n < \infty$. Prove that there are n points $x_1, x_2, \ldots, x_n$ in X such that the map $f \mapsto (f(x_1), \ldots, f(x_n))$ of V to $\mathbb{R}^n$ is an isomorphism.

Problem 17 (Sp88). 1. Let f be an analytic function that maps the open unit disc, $\mathbb{D}$, into itself and vanishes at the origin. Prove that

$$|f(z) + f(-z)| \leqslant 2|z|^2 \quad \text{in} \quad \mathbb{D}$$

2. Prove that the inequality in Part 1 is strict, except at the origin, unless f has the form $f(z) = \lambda z^2$ with λ a constant of absolute value one.

Problem 18 (Sp88). For which positive integers n is there a 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

with integer entries and order n; that is, $A^n = I$ but $A^k \neq I$ for $0 < k < n$? Note: See also **??**.

Problem 19 (Sp88). Show that one can represent the set of nonnegative integers, $\mathbb{Z}_+$, as the union of two disjoint subsets N_1 and N_2, such that neither N_1 nor N_2 contains an infinite arithmetic progression.

Problem 20 (Sp88, Sp10, Sp17). Does there exist a continuous real valued function $f(x)$, $0 \leqslant x \leqslant 1$, such that

$$\int_0^1 x f(x)\,\mathrm{d}x = 1 \quad \text{and} \quad \int_0^1 x^n f(x)\,\mathrm{d}x = 0$$

for $n = 0, 2, 3, 4, \ldots$? Give an example or a proof that no such f exists.